Aryan Padmakar Deshmukh

Software Engineer / Go / Ruby on Rails / React / AWS

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Summary

Software engineer building and running a production IoT metering and utility-billing platform in **Go**, **Ruby on Rails** and **React** on **AWS**. I own work end to end: MQTT ingestion, billing logic, admin UIs, releases and incident root-causing. Shipped **30+ pull requests** across 4 repositories in 9 months; converted from intern to full-time in 6.

Experience

Josh Software *IoT water-metering & utility-billing platform*

Pune, India | Jan 2026 – Present

Associate Software Engineer — Jul 2026 – Present

- **Billing correctness:** found bill due dates anchored to when an admin clicked *Approve* instead of the billing period, leaving **129 of 140** bills wrong in one cycle; made it a pure function of the period, pinned by specs.
- **Revenue leaks:** usage above the top tariff slab was silently unbilled (reproduced on a real **32,682-unit** bill); fixed it for water and electricity with API validation, and stopped new flats being billed before onboarding.
- **Phantom penalties:** root-caused late fees charged to residents who had already paid, correcting an earlier wrong diagnosis via read-only production queries; shipped an idempotent fix for all 4 cases.
- **Bulk upload pipeline:** React upload screens with browser-to-S3 presigned uploads and Sidekiq processing, reused across 3 import flows; cut device-service calls from one *per row* to **one per upload**, with a 409 concurrency guard.
- **Infrastructure & incidents:** built staging for the Go/MQTT service on AWS EC2 (systemd, Docker, nginx), fixing 6 latent bugs; diagnosed Postgres exhaustion (**100/100**) and a **408-job** Sidekiq backlog.
- **Go ingestion & releases:** Go ingestion now drops MQTT readings for inactive sites (React toggle, one query path across **10** call sites); cut production releases across 4 repositories.

Software Engineer Intern — Jan 2026 – Jun 2026

- **Negative water readings:** traced them to MQTT QoS 2 redelivery and server-side timestamps reordering events in the Go pipeline; moved to QoS 0 with device timestamps, after which these **support tickets stopped**.
- **Meter health:** Go cron job emailing weekly non-responsive-meter reports through AWS SES, plus Rails daily reports bucketing unreachable meters by 24 h and 3+ days.
- **Billing & payments:** consolidated duplicated penalty logic into one site-configurable component, fixed arrears carried forward after payment, and integrated a bank payment gateway in React.
- **Data correction:** admin API correcting consumption via offsets, replacing manual Rails-console fixes.

InveeSync *Frontend Developer Intern*

Remote | Dec 2024 – Jan 2025

- React/Redux workflows for a manufacturing-automation SaaS, with client-side caching to cut redundant API calls.

Projects

Smart Rule-Based Approval Engine *Go, Gin, PostgreSQL, React, TypeScript* — Live | GitHub

- Rules engine auto-deciding leave, expense and discount requests; Go cron expiry job, JWT roles, React UI.

Fintech Fraud Detection *Ruby on Rails, PostgreSQL, Solid Queue* — GitHub

- Real-time transaction risk scoring with device fingerprinting; fraud alerts and weekly summaries via Solid Queue jobs.

Technical Skills

Languages	Go, Ruby, C++, JavaScript, TypeScript, SQL
Backend & Data	Ruby on Rails, Gin, REST APIs, Sidekiq, MQTT, RSpec; PostgreSQL, TimescaleDB, MongoDB, Redis
Frontend	React, Redux-Saga, Zustand, TanStack Query, Next.js
Cloud & DevOps	AWS (EC2, S3, SES), Docker, nginx, systemd, Linux, Git, GitHub

Writing, Speaking & Open Source

- **Medium: three** posts on production debugging, e.g. *The Water That Flowed Backwards* (MQTT ordering in Go).
- **Talks:** GopherCon & RubyConf India 2026 proposals. **Open source:** merged PRs to CopyQ (12k stars, C++/Qt) and ONEARMY/community-platform (1.4k stars, React); reviews with fix PRs on helm/helm (30k stars), one adopted.

Education

B.Tech, Computer Engineering (Honours in Cyber Security), KKWIEER, Nashik	**9.26**	2026
Class XII (HSC), Vikas High School, Bhandara	**91.67%**	2022
Class X (CBSE), Royal Public School, Bhandara	**93.00%**	2020

Achievements & Leadership

- **Only student selected on day one** of campus placements, by the first recruiter on campus (500+ candidates).
- Ranked in the **top 5** of the Computer Engineering department; **Vice President, CSI Student Chapter** (2024–25).
- Captain of the school basketball team (district runners-up); led an e-waste collection and recycling drive.